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FPGA Implementation of Sound Encryption System Based on Fractional-Order Chaotic Systems

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Abstract— This paper introduces design and FPGA implementation of sound encryption system based on a fractional-order chaotic system. Also, it presents the FPGA implementation of Tang, Yalcin, and Özoğuz fractional order chaotic systems. The Grunwald-Letnikov definition is used to generalize the investigated systems into the fractional-order domain. Also, the variation of parameters for each system is investigated against window size of GL definition. Xilinx ISE 14.5 is used to simulate the proposed design. Also, some hardware reduction techniques are applied to decrease hardware utilization and increase throughput. Moreover, Yalcin system is employed as a chaotic generator in a speech encryption algorithm. Security analysis techniques are presented to show the robustness of the proposed algorithm. They include basic perceptual and statistical aspects, the NIST tests, key space analysis, mean square error (MSE) and key sensitivity. Statistical analyses including correlation, histogram, spectrogram, and entropy analysis are presented. The system is implemented on Artix-7 FPGA with one clock latency. Finally, experimental results are illustrated.

Keywords—*Fractional calculus; Chaos; Speech encryption; Chua; FPGA.*

I. INTRODUCTION

Data security has become a challenging field, and it is widely used on the internet to protect user's information [1]. Also, organizations and individuals commonly use encryption to protect sensitive data stored on computers and clouds [2]. Two different schemes for communication security can be used, which are symmetric encryption and asymmetric encryption. In symmetric encryption, the same secret key is used for encryption and decryption [3]. Asymmetric encryption is a type of security where two different keys are used, one for encryption and the other for decryption [4]. There are many cryptographic algorithms such as Data Encryption Standard (DES) [5], Advanced Encryption Standard (AES) [6], and Rivest–Shamir–Adleman (RSA) [7]. The main problem of these standards is that they have low-level efficiency and need a complex implementation which is not suitable for low-power applications. On the other hand, chaos-based cryptography can provide a fast and secure means for data protection. This is important for secure communication channels and multimedia protection [8].

The tremendous growth of mobile communications attracts more attention to the field of speech encryption to prevent attacks by eavesdroppers [9]. Attackers always make a huge effort to capture and decrypt the exchanged data between communication networks and misuse it. As a result, speech communication requires a higher level of security. Chaotic systems can be involved in digital speech encryption along with algorithms to give a powerful solution [10], [11], [12].

Chaos theory studies non-linear dynamical systems that are highly sensitive to initial conditions [13]. This behavior attracted many researchers to develop different models of chaotic systems. Chaotic systems are involved in many fields such as neural networks [14], economics [15] and communication security [9]. Encryption algorithms need robust random number generators to hide the information, and chaotic systems can fulfill this role [16],[17], [18]. Moreover, encryption algorithms need a large key space to prevent brute force attacks [19]. This can be guaranteed using chaotic systems with several

parameters. Also, chaotic systems can be generalized to fractional order domain using fractional calculus.

The fractional calculus is considered as a generalization of integer calculus. However, conventional calculus was used more often than the fractional one because of the absence of a numerical solution for it [20]. Fractional calculus adds extra degrees of freedom to the system which are the fractional orders. This enhances the controllability of the system and makes it more difficult to be hacked. The main reason for the wide usage of the integer order models was the absence of solution methods that solve the fractional order models. However, recently many definitions came out such as Grunwald–Letnikov (GL), Riemann–Liouville (RL) and Caputo definitions [20],[21]. These definitions made the fractional calculus has an important role in many applications such as bioengineering [22], electromagnetic waves [23], electrical engineering [24],[25],[26], filters[27],[28], robotics [29] and chaos theory [30].

In this work, GL definition mentioned in [19] is used, and it can be written as follows:

$${}^{\text{GL}}D_t^q f(t) = \lim_{h \rightarrow 0} \frac{1}{h^q} \sum_{j=0}^{\left\lfloor \frac{(t-a)}{h} \right\rfloor} w_j^{(q)} f(t - jh), \quad (1)$$

where $w^{(q)}$ is the binomial coefficient for fractional order q , defined as:

$$w_0^{(q)} = 1, \quad w_j^{(q)} = \left(1 - \frac{q+1}{j}\right), j = 1, 2, 3.. \quad (2)$$

The digital implementation of GL was introduced in many research articles to realize the fractional order differentiator and integrator [20], [31],[32], [33],[34]. In [31], Finite Impulse Response (FIR) filter was proposed to implement the fractional order operator. Short memory method was used to implement the fractional order differentiator and integrator on FPGA in [20]. The GL fixed window approach was introduced to implement the fractional order derivative in [32]. Moreover, it was used to implement different fractional-order chaotic systems with various window size [33]. Another attempt to implement the fractional order integrator and differentiator was presented in [34]. The implementation was built based on linear, quadratic and Piece-Wise Linear PWL approaches.

The digital implementation of fractional order chaotic systems has shown a noticeable interest in the last few years [35],[33]. Many chaotic systems were modeled with fractional calculus and implemented on FPGA such as Lorenz, Chen, Lu, and Chua systems [36]. Also, implementation of different families of multi scroll fractional-order chaotic oscillators such as 1-D 6 scrolls, 2-D 6 scrolls and 3D 6×6×6 scrolls using FPGAs and GL definition was discussed in [37]. Involving chaotic systems with encryption algorithms provided a new powerful method for cryptography. Some of this work was digitally realized and employed in many applications such as image encryption [16]. Speech encryption was carried out by [38] using Rossler and Chen systems on Virtex family FPGA. Also, text and image encryptions using Lorenz chaotic system were implemented on FPGA by [39]. Moreover, [40] presented a data stream encryption using Lu chaotic system as a key generator with FPGA maximum clock frequency of 22.868 MHz. In [41], speech encryption system was implemented based on the modified chaotic logistic map and bit permutations. The FPGA implementation guaranteed low power consumption, high parallelism, and good hardware. Furthermore, combining chaotic systems with encryption algorithms to be realized on hardware was difficult without using digital implementation such as FPGAs.

The main contribution of this work is the implementation methodology of three fractional chaotic systems: Tang, Yalcin and Özoğuz systems with an 8-scrolls chaotic attractor. These implementations are used as a random generator for an asymmetric encryption system. The system's behavior with different window sizes and performances on FPGA are investigated.

Moreover, the proposed speech encryption scheme and its FPGA implementation are introduced. Finally, many speech analysis techniques are followed by the proposed encryption system to ensure its robustness against different attacks. The analyses are key space analysis, mean square error (MSE), key sensitivity and statistical analysis. Statistical analyses include correlation, histogram, spectrogram and entropy analysis.

This paper is organized as follows; section II introduces the investigated chaotic systems. Section III presents the design and implementation of the speech encryption system. Moreover, section IV investigates the speech security analysis, and section V is for discussion and results. Finally, section VI concludes all the work.

II. INVESTIGATED CHAOTIC SYSTEMS

Chua chaotic system was the first system to be developed by an electronic circuit inspired by Lorenz attractor [42]. Several modifications were applied to the simple Chua's circuit and other systems like EL-Wakil [43] to produce multi-scrolls. This was achieved by modifying the non-linear function of the chaotic system, to be more complex. In this section, a modification on Chua's circuit made by Tang is presented. Another two systems were introduced by Yalcin and Ozoguz based on El wakil simple double scroll system.

A. Tang system theoretical analysis

To generate multi scrolls chaotic system, Tang [44] modified the simple Chua's circuit shown in Fig.1 (a). This system can be described as follows:

$$\dot{x} = \alpha(y - f(x)), \quad (3a)$$

$$\dot{y} = x - y + z, \quad (3b)$$

$$\dot{z} = -\beta y, \quad (3c)$$

where $f(x)$ represents the piecewise-nonlinear function given by:

$$f(x) = \begin{cases} \frac{b\pi}{2a}(x - 2ac), & x \geq 2ac, \\ -b \sin\left(\frac{\pi x}{2a} + d\right), & -2ac < x < 2ac, \\ \frac{b\pi}{2a}(x + 2ac), & x \leq -2ac, \end{cases} \quad (4)$$

where α, β, a, b, c and d are system parameters. The system generates N scrolls with N equals:

$$N = c + 1, \quad (5)$$

and,

$$d = \begin{cases} \pi & \text{if } c \text{ is even,} \\ 0 & \text{if } c \text{ is odd.} \end{cases} \quad (6)$$

The non-linear function presented in (4) is depicted in Fig.1 (b). The GL definition (1) requires a very large memory which is not practical. It was recommended by many researchers to limit the number of terms calculated in this summation as this series is not changed after a certain number of terms.

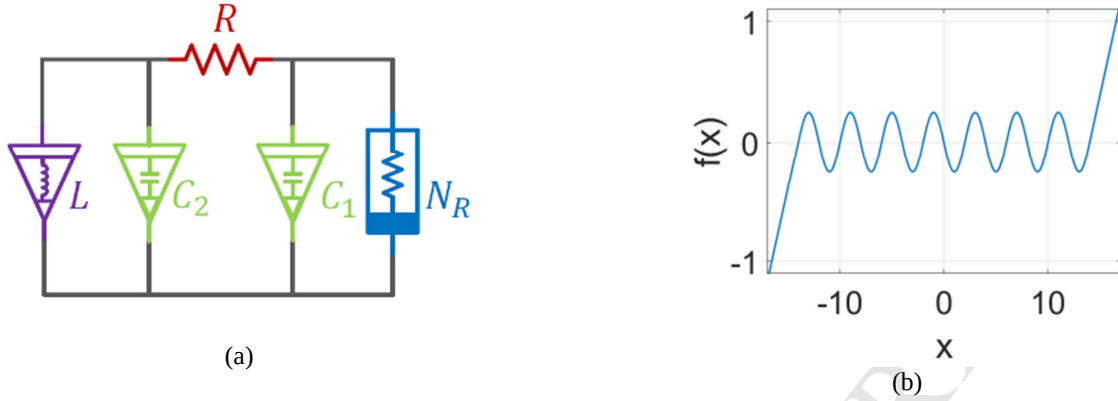


Fig. 1. a) Chua circuit, and b) The non-linear function $f(x)$ for $a = 1$, $b = 0.25$ and $c = 7$.

Applying GL definition on (3); the fractional order Tang's system can be written as follows:

$$x_{t_{n+1}} = \alpha h^{q_1} (y_{t_n} - f(x)) - \sum_{j=1}^L w_j^{(q_1)} x(t_n - j), \quad (7a)$$

$$y_{t_{n+1}} = h^{q_2} (x_{t_n} - y_{t_n} + z_{t_n}) - \sum_{j=1}^L w_j^{(q_2)} y(t_n - j), \quad (7b)$$

$$z_{t_{n+1}} = \beta h^{q_3} (-y_{t_n}) - \sum_{j=1}^L w_j^{(q_3)} z(t_n - j), \quad (7c)$$

where h is the step size, L is the window size and q_1, q_2 and q_3 are the fractional orders. Figure 2 depicts numerical simulations by MATLAB for fractional order Tang system with 8-scrolls chaotic attractors. The system parameters are chosen such that, $h = 2^{-6}$, $q_1 = q_3 = 0.94$, $q_2 = 0.85$, $\alpha = \beta = 16$, $a = 1.25$, $b = 0.25$ and $c = 7$.

B. Yalcin and Özoğuz's systems theoretical analysis

In this subsection, two multi-scroll chaotic systems based on El-Wakil system [43] are presented. One was firstly realized by Özoğuz's [45]; the target of this system is to be complex and consume less hardware than other investigated systems. This realization appeared to be simpler than Chua's circuit. Moreover, it has no inductors. The other system was introduced by Yalcin [46]. This system belongs to 1-D scroll grid attractors. Grid scroll attractor's family is divided into three subfamilies 1-D, 2-D, and 3-D. This classification is according to the number of non-linear functions in addition to the number of equilibrium points which determines the location and number of scrolls. These systems are distinguished by their few numbers of parameters and simple equations which reflects on the simplicity of implementation. Moreover, their outputs are very interesting and produce N controllable scrolls. The systems differential

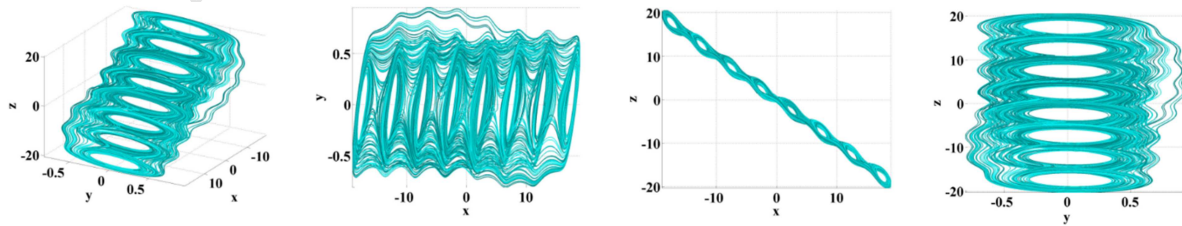


Fig. 2. The fractional order Tang 8-scrolls chaotic attractors (MATLAB results).

equations with their non-linear functions are shown in Table I. Also, both systems' chaotic attractors and their non-linear functions are shown in Table II using MAT-LAB with their parameters values to obtain 8-scrolls for both systems.

C. Parameters effects

The performances of the three studied systems are investigated by sweeping window size L with system parameters. For Tang's system, Table III shows the change of the non-linear function parameter a with the window size L . The parameters of the nonlinear function a and b have the same effect but b has a smaller range than a . From Table III, it was concluded that the sensitivity of this system comes from its nonlinear function mostly.

Özoğuz's system which introduced the tanh approach to El-Wakil systems, has many variables that control number of scrolls such as a , M , N and O^* where O^* is a pure imaginary number of parameter O . In Table IV, variations of system parameter a with window size L are illustrated to discuss how a and L control the resultant chaotic attractors. For the chaotic behavior, the range of a is approximately from 0.014 to 0.29.

For Yalcin's system, Table V shows variations of the parameter a with the window size L . As the window size L increases, the system performance will be improved and gives a larger number of scrolls. On the other hand, by increasing a from 0.1 to 0.9, the system starts to behave periodically and becomes non-chaotic. This concludes, increasing window size L will improve the output of the system, but also increase the hardware resources, as it will need more memory. Also increasing the value of a will lead the system to be periodic gradually as shown in Table V.

TABLE I. Systems equations.

System	Differential equation	$f(x)$	No. of scrolls
Özoğuz's modification (Tanh approach)	$D^{q1}x = y$ $D^{q2}y = z$ $D^{q3}z = -a(y + z + f(x))$	$f(x) = \sum_{j=-N}^M (-1)^{j-1} \tanh k(x - o_j),$	$\frac{M + N + 2}{2}$
Yalcin's modification (piece-wise nonlinear function)	$D^{q1}x = y$ $D^{q2}y = z$ $D^{q3}z = -a(-x - y - z + f(x))$	$f(x) = \sum_{i=1}^{M_x} g_{\frac{(-2i+1)}{2}}(x) + \sum_{i=1}^{N_x} g_{\frac{(2i-1)}{2}}(x),$ $g_{\theta}(3) = \begin{cases} 1 & 3 \geq \theta & \theta > 0 \\ 0 & 3 < \theta & \theta > 0 \\ 0 & 3 \geq \theta & \theta < 0 \\ -1 & 3 < \theta & \theta < 0 \end{cases}$	$N_x + M_x + 1$

TABLE II. Yalcin and Özoğuz's systems attractors and their non-linear function.

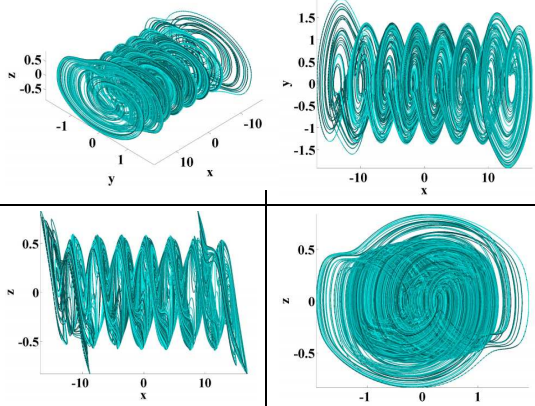
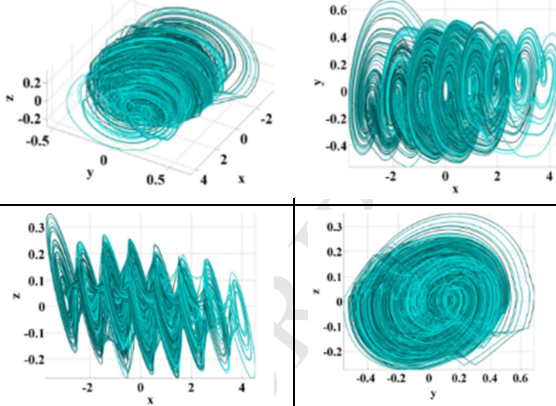
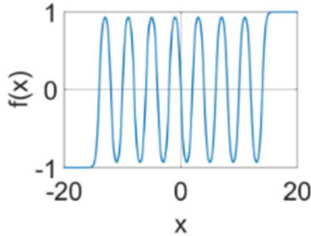
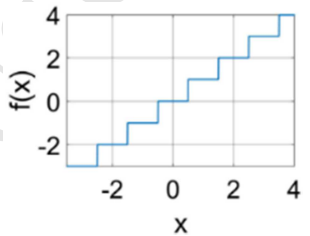
	Özoğuz's modification (Tanh approach)	Yalcin's modification (piece-wise nonlinear approach)
chaotic attractors		
Non-linear function		
Value of parameters	$h = 0.09, M = 7, N = 7, q_1 = q_2 = q_3 = 0.982, a = 0.25$ and $o_j = 2j$	$a = 0.21, q_1 = q_2 = q_3 = 0.98, N_X = 4$ and $M_X = 3$

TABLE III. the variation of window size with system parameters for Tang system.

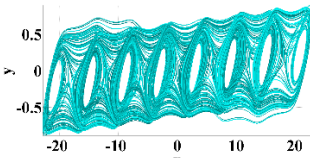
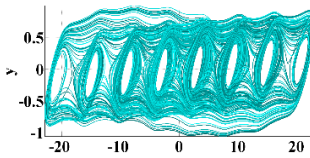
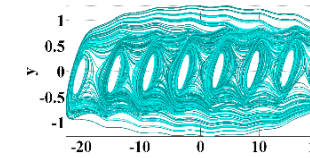
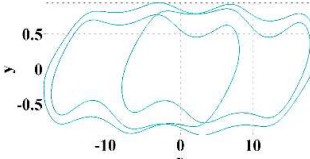
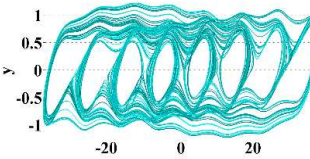
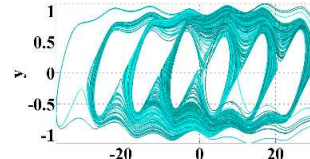
a	L=20	L=30	L=40
1.5			
2.5			

TABLE IV. the variation of window size with system parameters for Ozugos system.

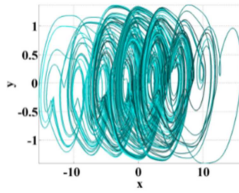
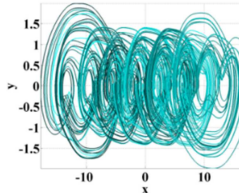
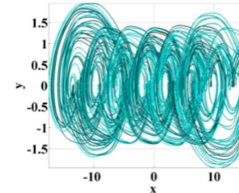
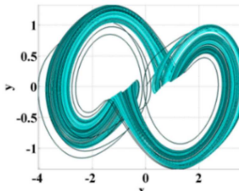
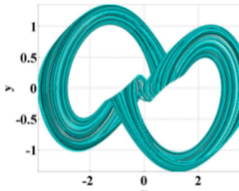
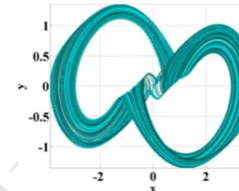
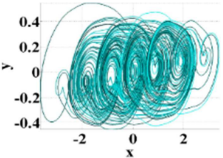
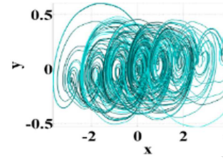
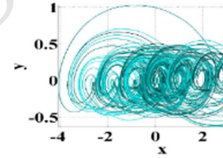
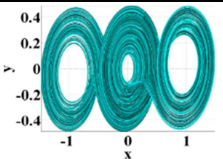
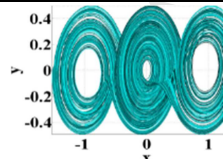
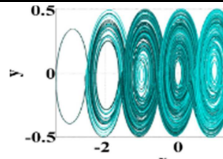
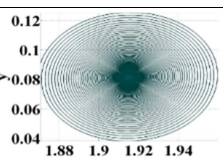
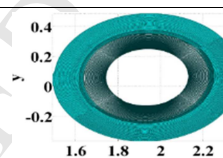
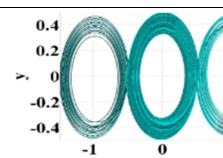
a	$L = 11$	$L = 40$	$L = 200$
0.014			
0.29			

TABLE V. the variation of window size with system parameters for Yalcin system.

a	$L = 15$	$L = 20$	$L = 35$
0.1			
0.7			
0.9			

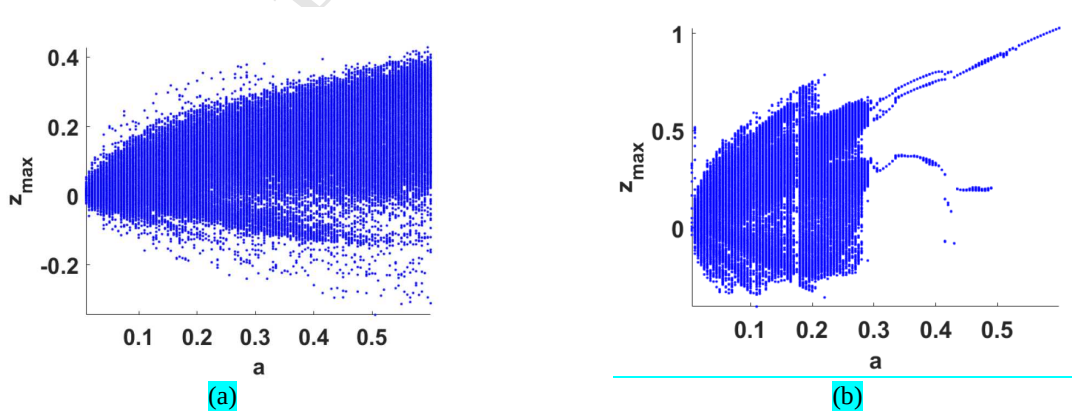


Fig. 3. Bifurcation diagram of, (a) Yalcin's system for the parameter a , and (b) Özoğuz's system for the parameter a

Bifurcation diagrams define the parameters range corresponding to chaotic responses. Figure 3 presents the bifurcation diagram of the proposed chaotic systems, from which parameter ranges

corresponding to chaotic behavior can be expected. The time series show whether the solution is stable, periodic or chaotic. Figure 3 (a) and 3(b) presents the bifurcation of Yalcin's system, and Özoğuz's system for the parameter a , respectively.

D. FPGA implementation

The studied systems are implemented using three registers to store the previous values of the variables x , y and z besides three blocks called numerical solution evaluates each variable based on other variables values. These three registers are used to compute the updated versions of x , y and z values as illustrated in Fig. 4. In the presented systems, the values of constants are chosen to be multiple of 2. Any multiplication in the three differential equations can be replaced by just a simple shift left or write operation. For the Tang system as shown in Fig. 5, the Shift Arithmetic Right (SAR) value represents the multiplication of αh^{q_1} , h^{q_2} and βh^{q_3} in the main differential equations. In the hardware implementation, non-linear functions are troublesome issues. To reduce complexity, the curve fitting technique is used to reduce the complexity. Curve fitting is a numerical mathematical technique used to approximate functions to another mapped relations. After curve fitting, hyperbolic tangent function and sine function in Özoğuz's and Tang's system followed other piecewise relations define by $T(x)$ and $S(x)$ respectively. The approximate sine function $S(x)$ with piecewise relations is shown in Table VI. The approximate hyperbolic tangent function can be described as follows:

$$T(x) = \begin{cases} -1, & -2\pi \leq x < -1, \\ x, & -1 \leq x \leq 1, \\ 1, & -1 < x \leq 2\pi. \end{cases} \quad \begin{matrix} (8a) \\ (8b) \\ (8c) \end{matrix}$$

Also, both $S(x)$ and $T(x)$ are compared to their correspondents time continuous functions as shown in Table VI.

Figure 6 shows the proposed block diagram of 8-scrolls Özoğuz fractional system on FPGA with the tanh function as a non-linear function designed with curve fitting techniques. Also, Fig.7 proposes the block diagram of 8-scrolls Yalcin fractional system on FPGA with non-a linear function designed with PWL approach. Five logic utilizations were chosen to compare the systems which are: number of employed registers, number of the look-up tables (LUT), clock speed, throughput and number of Digital Signal Processing blocks (DSP). The performance of the three systems is shown in Table VII. From the results, the second system which is the tanh approach is found to use the biggest resources which cause a big delay and the smallest throughput. On the other hand, Yalcin and Tang systems have approximately the same resources. Yalcin's system has better throughput than the Tang system. This happens because of the sine function in Tang's system; which consumes more resources than the piecewise linear (PWL) function used by the Yalcin's system.

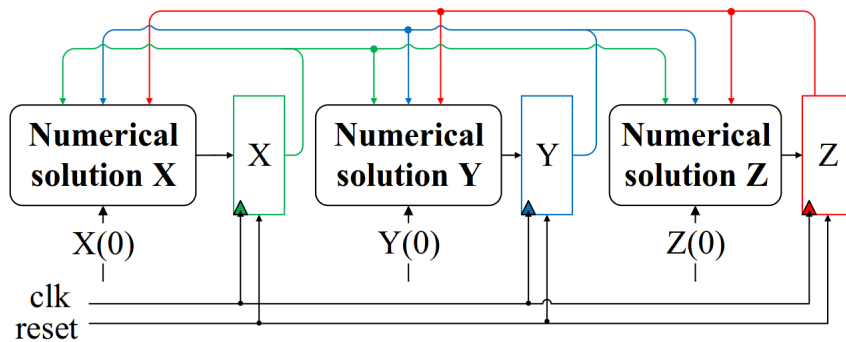


Fig. 4. The proposed general block diagram of three differential equations implementation on FPGA.

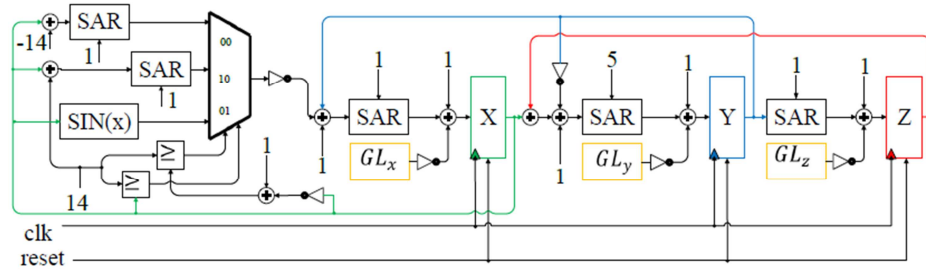


Fig. 5. The proposed block diagram of 8-scrolls Tang fractional system on FPGA with sine approach.

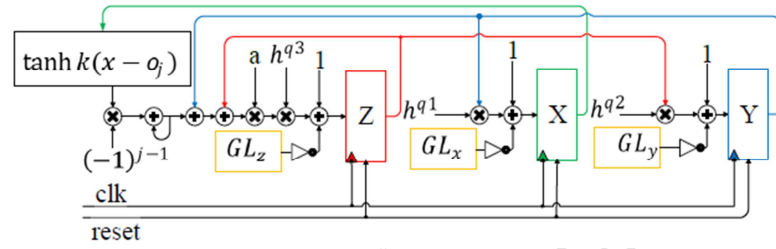


Fig. 6. The proposed block diagram of 8-scrolls Özoğuz fractional system on FPGA with tanh approach.

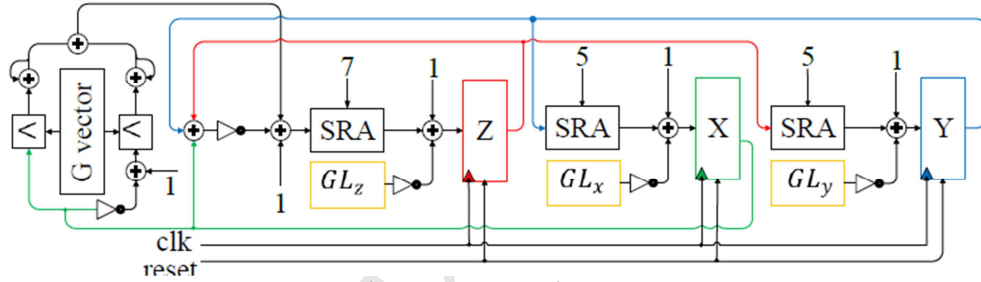


Fig. 7. The proposed block diagram of 8-scrolls Yalcin fractional system on FPGA with PWL approach.

TABLE VI. Curve fitting of sine and tanh functions, the results compared to continuous time functions.

Region	$S(x)$	Region	$S(x)$	Region	$S(x)$
$S(x)$ Vs $\sin(x)$			$T(x)$ Vs $\tanh(x)$		
$0 \leq x < \frac{\pi}{6}$	$x - 0.01$	$\frac{\pi}{1.618} \leq x < \frac{\pi}{1.3334}$	$-0.5x + 1.895$	$1.3820\pi \leq x < \frac{3\pi}{2}$	$-0.125x - 0.4$
$\frac{\pi}{6} \leq x < \frac{\pi}{4}$	$0.75x + 0.12$	$\frac{\pi}{1.3334} \leq x < \frac{\pi}{1.2}$	$-0.75x + 2.46$	$\frac{3\pi}{2} \leq x < 1.6180\pi$	$0.125x - 1.59$
$\frac{\pi}{4} \leq x < \frac{\pi}{2.618}$	$0.5x + 0.33$	$\frac{\pi}{1.2} \leq x < \frac{7\pi}{6}$	$-x + 3.15$	$1.6180\pi \leq x < \frac{7\pi}{4}$	$0.5x - 3.47$
$\frac{\pi}{2.618} \leq x < \frac{\pi}{2}$	$0.125x + 0.79$	$\frac{7\pi}{6} \leq x < \frac{5\pi}{4}$	$-0.75x + 2.23$	$\frac{7\pi}{4} \leq x < 1.8333\pi$	$0.75x - 4.83$
$\frac{\pi}{2} \leq x < \frac{\pi}{1.618}$	$-0.125x + 1.2$	$\frac{5\pi}{4} \leq x < 1.3820\pi$	$-0.5x + 1.24$	$1.8333\pi \leq x \leq 2\pi$	$x - 6.25$

TABLE VII. Comparison between three investigated chaotic systems.

System	No. of registers	No. of LUT	DSP block	Max. Clock (MHz)	Throughput (Gb/s)
Tang approach	1,763 (1%)	3,339 (5%)	134 (55%)	67.378	6.198 (for 96 bit output)
Özoğuz approach	2597 (2%)	3856 (6%)	194 (80%)	42.052	4.036 (for 96 bit output)
Yalcin approach	1932(1%)	2125(3%)	132(55%)	94.941	9.114 (for 96 bit output)

III. SPEECH ENCRYPTION

In this section, a speech encryption algorithm is introduced based on stream ciphering blocks. The encryption blocks have been chosen to add complexity without consuming a lot of hardware resources. The introduced blocks are, Substitution Box (S-BOX), Permutation, Rotation, XOR operation, and feedback. The sequence of the blocks has been chosen by testing most of the possibilities and get their correlation with the input speech signal. The used one has the lowest correlation between the other five sequences. The block diagram of encryption and decryption systems are shown in Fig. 8(a) and Fig. 8(b), respectively. Also, Fig.9 (a) shows the flow of the encryption process in addition to the system's inputs and output. The speech sample used is a standard sample: "We shall fight on the beaches - The International Churchill Society". The following subsections will discuss each encryption block separately as well as the FPGA implementation of the encryption system.

A. Feedback block

Cipher block chaining mode (CBC) is used to make the feedback of the output. This allows output not to depend only on the current input but also on previous outputs. The design detailed block diagram is shown in Fig. 9(b). The encryption equation of CBC can be written as follows:

$$y_i = e_K(M_i \oplus y_{i-1}), \quad (9)$$

where

$$y_1 = e_K(M_1 \oplus a), \quad (10)$$

and the decryption equation will be:

$$M_i = e_K^{-1}(y_i) \oplus y_{i-1}, \quad (11)$$

where

$$M_1 = e_K^{-1}(y_1) \oplus a. \quad (12)$$

B. Substitution Box block

The function of this block is to substitute each byte with unrelated byte value to confuse the input stream. The employed S-Box was introduced in [47]. Table VIII shows the routing table and operation for S-box.

C. Permutation block

The permutation is another method for distraction and is done on a bit level. This technique is done by mapping single bits to different locations based on four permutation schemes. This technique selects different four permutation schemes as shown in Table IX based on the two least significant bits (LSBs) of the state variable X of the chaotic generator

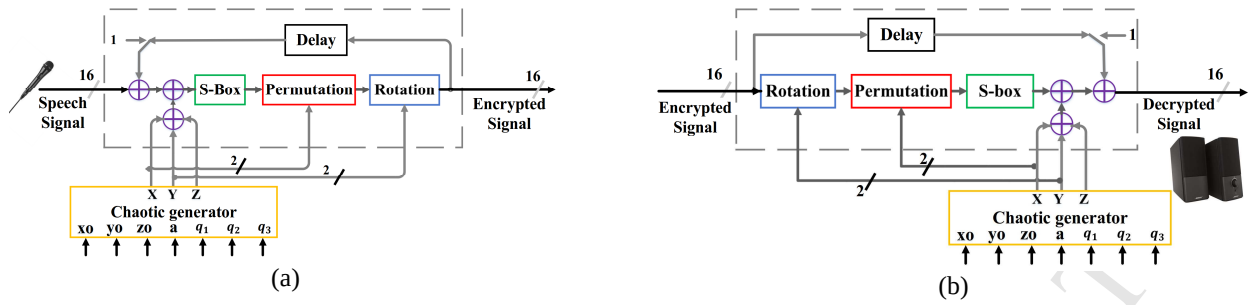


Fig. 8. a) Encryption block diagram, and b) Decryption block diagram.

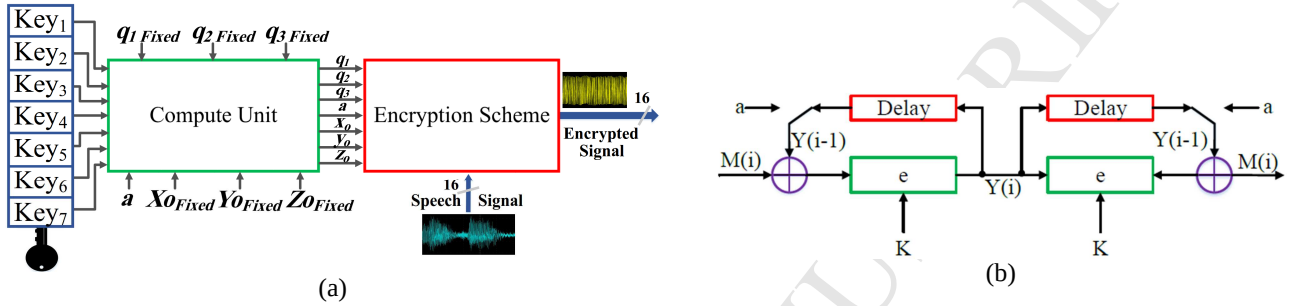


Fig. 9. a) The layout of the Encryption process, and b) Cipher block chain (CBC) block.

D. Rotation block

As the permutation block, four rotational bit schemes are utilized based on the Rotate Right operation (RoR). The two LSBs of state variable Y of the chaotic generator determine the used scheme. Table X shows the operation of the rotation block.

E. Chaotic generator

From the previous section, a comparison is made between the three 8-scrolls chaotic systems. Yalcin system has shown the highest throughput on FPGA with acceptable chaotic behavior. As a result, Yalcin has been chosen to be the chaotic, random number generator of the cryptosystem.

After applying the previous blocks in the proposed sequence, the results from the encryption system on the standard speech mentioned before is shown in Fig. 10. The results show that the original speech is perfectly encrypted and became white noise.

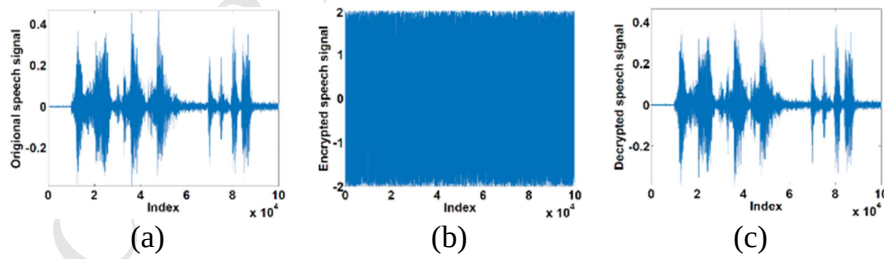


Fig. 10. a) Original speech, b) Encrypted speech and c) Decrypted speech.

TABLE VIII. S-Box block routing table.

S-Box																
x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$S(x)$	C	5	6	B	9	0	A	D	3	E	F	8	4	7	1	2

TABLE IX. Permutation block.

1 st Permutation									3 rd Permutation								
I/P Bit index	0	1	2	3	4	5	6	7	I/P Bit index	0	1	2	3	4	5	6	7
O/P Bit index	14	8	10	0	12	1	13	3	O/P Bit index	14	12	5	1	11	8	10	6
I/P Bit index	8	9	10	11	12	13	14	15	I/P Bit index	8	9	10	11	12	13	14	15
O/P Bit index	11	15	6	9	2	4	7	5	O/P Bit index	13	4	3	15	0	2	9	7
2 nd Permutation									4 th Permutation								
I/P Bit index	0	1	2	3	4	5	6	7	I/P Bit index	0	1	2	3	4	5	6	7
O/P Bit index	1	11	13	2	10	14	12	5	O/P Bit index	5	14	11	4	13	10	7	15
I/P Bit index	8	9	10	11	12	13	14	15	I/P Bit index	8	9	10	11	12	13	14	15
O/P Bit index	4	15	7	3	9	8	0	6	O/P Bit index	2	12	9	0	3	1	6	8

TABLE X. Rotation block

ROTATION			
1 st Rotation	2 nd Rotation	3 rd Rotation	4 th Rotation
>>5	>>1	>>9	>>3

TABLE XI. FPGA summary for each encryption block.

	Hardware resources	Maximum propagation delay
Xoring and Feedback Blocks	Number LUT: 16	0.934 ns
Permutation Block	16× 4 multiplexer: 1	1.139 ns
S-Box Block	Number of LUT: 16	0.948 ns
Rotation Block	16× 4 multiplexer: 1	1.139 ns

F. FPGA implementation

The digital implementation of the presented encryption scheme is straightforward with low-level hardware complexity. Table XI illustrates both the hardware resources and maximum propagation delay for each encryption block. The presented encryption system requires few hardware resources with low propagation delays which is suitable for real-time applications. The analog speech is sampled with 16 bits per sample as well as the speaker output.

For this reason, truncation has done for Yalcin chaotic generator outputs which were 32-bits each. The encryption design is based on two levels of mapping, byte and bit levels. S-Box is the block that is responsible for byte mapping. XOR, permutation, and rotation blocks are responsible for bit mapping.

IV. SPEECH SECURITY ANALYSIS

This section will discuss the performance of the encryption system. Several analyses are used to test the system as key space analysis, mean square error (MSE), key sensitivity and statistical analysis. Statistical analyses include correlation, histogram, spectrogram and entropy analysis.

A. Keyspace analysis

Brute force attack is a trial and error method where the attacker tries all possibilities of the system's key. This type of hacking is nonsense in case of large key space. One of the main objectives for any cryptosystem is to have a large key space. The keyspace is the number of encryption keys that are available in a cryptosystem. According to Advanced Encryption Standard (AES), the minimum number of bits for the encryption key are 128 with 2^{128} Possibilities.

In this subsection, Keyspace is calculated using Xilinx ISE 14.6. Two different Yalcin systems were presented to obtain 8-scrolls. The Yalcin's parameters chosen in the earlier sections which have equal fractional orders is a bad choice for encryption. As this system has only five different parameters with 32-bit representation for each (8 bits for the integer part and 24 bits for the fractional part). So the maximum key space for this system will be 2^{120} Which is below standard. To increase keyspace, a modification on the Yalcin system was made to make the fixed values of the new fractional-order as 0.978, 0.982 and 0.981 for q_1 , q_2 and q_3 respectively.

The chaotic behavior of the new system is similar to Yalcin's system discussed earlier. To calculate the key space, tiny changes in Yalcin parameters are added. The decrypted signal deviates from the original speech in case of perturbations in system parameters: $x_o, y_o, z_o, a, q_1, q_2$ and q_3 with $2^{-24}, 2^{-24}, 2^{-24}, 2^{-20}, 2^{-21}, 2^{-21}$ and 2^{-19} respectively as illustrated in Table XIV. This will give a key space of 2^{153} which is above standard. Figure 11 shows the generation of the key used for both encryption and decryption systems. The key consists of seven sub-keys match seven parameters. Each sub-key is concatenated with zero's then added to the fixed values to compute the chaotic parameters. The sub-keys include different numbers of bits, where $key_1, key_2, \dots$ and key_7 required 21, 23, 23, 21, 20, 20 and 21 respectively. As a result, the actual key space will be represented with 149 bits not 153 as mentioned before and this still above AES.

B. Sensitivity

The sensitivity of the investigated decryption system to small perturbations in encryption key could be estimated by Mean Square Error (MSE) and entropy tests. MSE is a mathematical tool used to test the efficiency of an encrypted system. This is achieved by describing the deviation of the encrypted signal from the original one. Also for the wrong key, MSE is used to deviations of wrong decrypted signals from the original one.

MSE is calculated as follows [48]:

$$MSE = \frac{1}{N} \sum_{i=1}^N (S(i) - E(i))^2, \quad (14)$$

where N is the length of the original and encrypted signals, S is the original speech signal and E is the encrypted signal. In many cases, the design aims to minimize MSE but in encryption systems, the higher the value of MSE, the better performance of the system. However, encryption systems aim to minimize MSE as possible. Calculated MSE for the presented encryption system is 3.577×10^8 for 100,000 samples of the standard used speech file.

Entropy analysis is used to measure the randomness of the ciphered data. It is used to prove the unpredictability of encrypted data and how much it is close to the white noise. The more the value of the entropy approaches the number of bits per symbol, the more the encrypted data is considered random. So in our case, the ideal value for the entropy is equal to 16. The Entropy can be calculated by the following expression [49]:

$$Entropy = - \sum_{i=1}^{2^B} P(S_i) \times \log_2 P(S_i), \quad (15)$$

where $P(S_i)$ is the probability distribution function of the input speech data and B is the symbol length which is in our case is 16 bits. For 100,000 samples of the speech signal, the entropy of the encrypted signal is 15.45452.

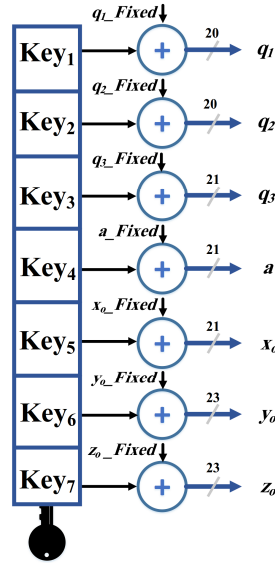


Fig. 11. Key generation of the encryption system.

C. Statistical analysis

The correlation coefficient is one of the most commonly used parameters in statistical analyses. This type of analyses is used to estimate the relationship between the original input signal and the ciphered signal. The correlation coefficient (r) formula is given by [4]:

$$Cov(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \frac{1}{n} \sum_{j=1}^n x_j) (y_i - \frac{1}{n} \sum_{j=1}^n y_j), \quad (16a)$$

$$D(x) = \frac{1}{n} \sum_{i=1}^n \left(x_i - \frac{1}{n} \sum_{j=1}^n x_j \right)^2, \quad (16b)$$

$$r = \frac{Cov(x, y)}{\sqrt{D(x)} \sqrt{D(y)}} \quad (16c)$$

where n is the length of both signals x and y . Perfect correlation ($r = 1$) means a strong relationship between the two signals. Zero correlation indicates large deviations between x and y . The correlation coefficient is found to be: 0.001439 for 100,000 input speech signal samples.

The histogram is used to estimate the probability distribution of a variable. As the distribution of the signal becomes equal, it becomes harder to detect the transferred data. In Table XII, the histogram is calculated for 100,000 samples of the input speech signal. The data is stored into 100 equally spaced bins for both speech and encrypted signals. The y-axis represents the number of elements per each bin. Results show the encrypted data is approximately equally distributed, so there is no certain samples have significantly large values than others.

Spectrogram analysis is a random test tool that divides the signal in the time domain into slots to calculate Fast Fourier Transform (FFT) for each slot. The magnitude square of FFT is plotted versus each slot to indicate the energy of sound. In Table XII for 100,000 samples of the input speech signal, the spectrogram of original and encrypted signals is made. The energy of sound appears to be small as a result of low speech signal's amplitude, so spectrogram is measured in logarithmic. For the original input speech, high energy sound components appear at low frequencies as expected, and after encrypting the speech file, spectrogram appears random which indicates how randomness is the distribution of sound components' energy.

For this reason, distinguishing the encrypted speech from the original speech signal is impossible. In Table XIII, the waveforms of the decrypted speech signal in case of small perturbations of the encryption key are shown. The decrypted signals show that the system is sensitive to tiny changes in the key and the decrypted signal doesn't relate to the original signal in case of a small perturbation. Also in Table XIV, several analyses are made for perturbations in the encryption key. Both MSE and correlation coefficient are calculated between the original speech and wrong decrypted signals. The results show that the MSE is very large and the correlation coefficient has the lowest values between original speech and wrong decrypted one. On the other hand, entropy is calculated for only the wrong decrypted signal in each case which has values near to ideal entropy.

TABLE XII. Histogram and spectrogram for both original speech and encrypted signals.

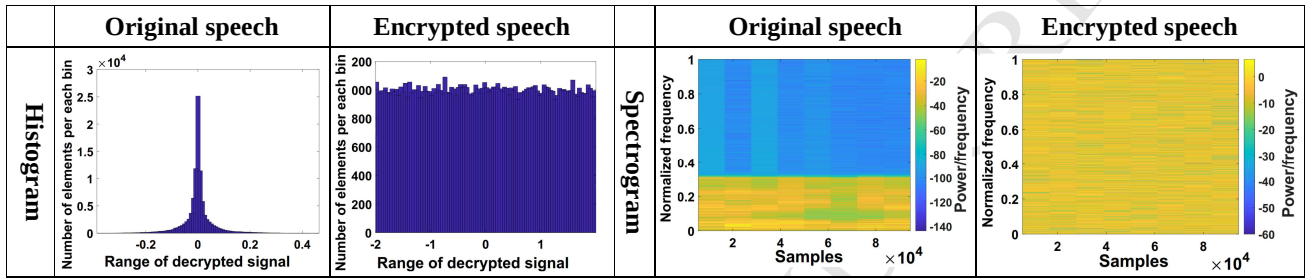


TABLE XIII. Decrypted speech signals in case of perturbations in the encryption key.

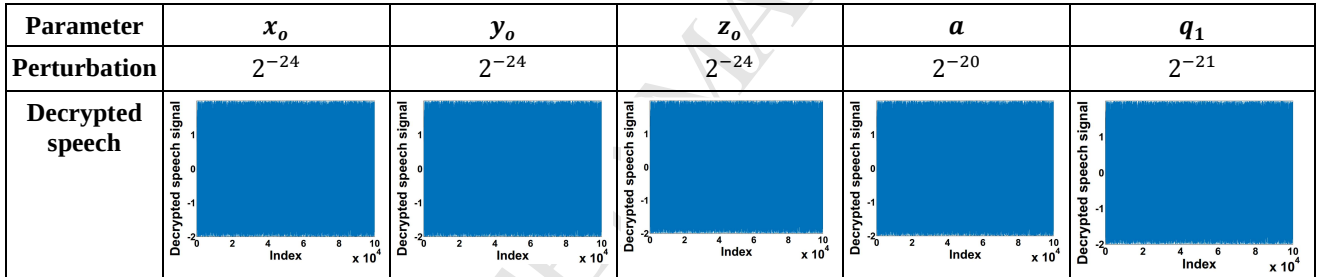


TABLE XIV. Analyses in case of perturbations in the encryption key.

Parameter	x_o	y_o	z_o	a	q_1	q_2	q_3
Perturbation	2^{-24}	2^{-24}	2^{-24}	2^{-20}	2^{-21}	2^{-21}	2^{-19}
MSE $\times 10^8$	3.53585	3.52969	3.5153	3.55608	3.53889	3.568	3.5644215
Entropy	15.4132	15.4215	15.4062	15.44129	15.432	15.442	15.442309
Correlation coefficient	-6.96×10^{-4}	-0.0019	-0.0022	-0.00538	-0.0026	0.0010	-4.144×10^{-4}

TABLE XV. NIST results for the encrypted speech signal.

TEST	P-VALUE	RESULT	PROPORTION	RESULT
Frequency	0.183323	✓	1.000	✓
BlockFrequency	0.637119	✓	1.000	✓
CumulativeSums	0.213309	✓	1.000	✓
Runs	0.066882	✓	1.000	✓
LongestRun	0.006196	✓	1.000	✓
Rank	0.534146	✓	1.000	✓
FFT	0.437274	✓	1.000	✓
NonOverlappingTemplate	0.353523	✓	0.990	✓
OverlappingTemplate	0.275709	✓	1.000	✓
Universal	0.090936	✓	1.000	✓
ApproximateEntropy	0.162606	✓	1.000	✓
RandomExcursions	0.292671	✓	0.990	✓
RandomExcursionsVariant	0.481085	✓	0.991	✓
Serial	0.223077	✓	1.000	✓
LinearComplexity	0.437274	✓	0.958	✓

NIST tests are used to measure the randomness properties of the encrypted speech signal. They are processed as a series of bits for which the P-value distribution (PV) and the proportion of passing sequences (PP) are estimated. Table XV presents that the encrypted speech signal file successfully pass the tests.

V. DISCUSSION AND RESULTS

In this section, the FPGA obtained results for chaotic systems are illustrated. The proposed systems are realized on Xilinx Artix 7 FPGA and simulated on Xilinx ISE 14.6.

A. Chaotic systems FPGA results

The phase portraits of three chaotic systems: Tang, Yalcin, and Ozoguz are depicted in Table XVI. The chaotic behaviors of Yalcin are illustrated for the original parameters where all fractional orders have the same value of 0.98. It is proved that the FPGA results are consistent with the MATLAB results in Table II. FPGA setup is shown in Fig.12 (a).

B. Encryption system FPGA Results

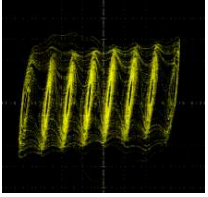
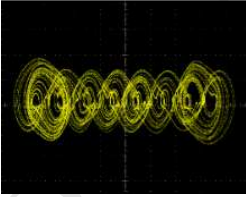
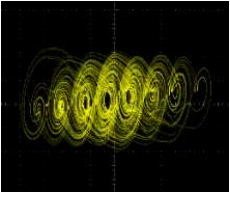
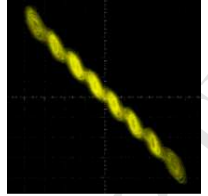
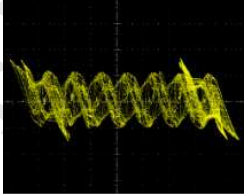
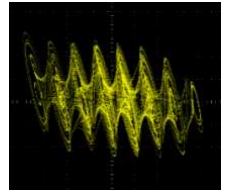
The encryption system is tested on FPGA and oscilloscope for speech signals. Figure 12 (c) shows the original sound signal (the lower signal) and the encrypted signal (the upper signal). It's clear that the encrypted signal has equal magnitude overall signal and can't be detected. Also, Fig.12 (b) shows the FPGA and oscilloscope setup.

VI. CONCLUSION

This paper introduced FPGA implementation of fractional order multi-scrolls Tang, Yalcin and Özoğuz chaotic systems. Grunwald-Letnikov's (GL) definition was used to generalize the chaotic system equations into the fractional-order domain. Also, the variation of parameters for each system was investigated against a window size of GL definition. Xilinx ISE 14.5 was used to simulate the

investigated design, and Artix-7 XC7A100T FPGA was used for systems realization. FPGA summary for all the fractional systems was introduced, and a comparison was made between them. The resulted throughput of each system was 3.417 GHz for Yalcin, 1.513 GHz for Özoğuz and 2.425 GHz for Tang. As a result, minimum throughput was for Özoğuz and maximum throughput for Yalcin so that Yalcin was chosen to be used in the speech encryption system. Moreover, Yalcin system was employed in the encryption algorithm for speech encryption. New encryption algorithm was proposed which is composed of some blocks such as permutation, S-Box, feedback, and chaotic generator. The blocks were arranged to get the best analyses results. Encryption analysis techniques were presented to show the robustness of the encryption algorithm. The analyses key space analysis, mean square error (MSE), key sensitivity and statistical analysis. Statistical analyses include correlation, histogram, spectrogram and entropy analysis. The entropy for 100,000 samples of encrypted speech signal was 15.45452. Also, key space by using seven the parameters of the chaotic generator was 149 bits which were higher than the AES standard. Also, calculated MSE for the encryption system was 3.577×10^{-8} for 100,000 samples of the standard speech file. The whole system was implemented on FPGA with one clock latency. Finally, a discussion on the results was made showing the output of three chaotic systems and encryption system.

TABLE XVI. FPGA phase portrait results for three chaotic systems.

	Tang system	Ozoguz system	Yalcin system
X-Y			
X-Z			

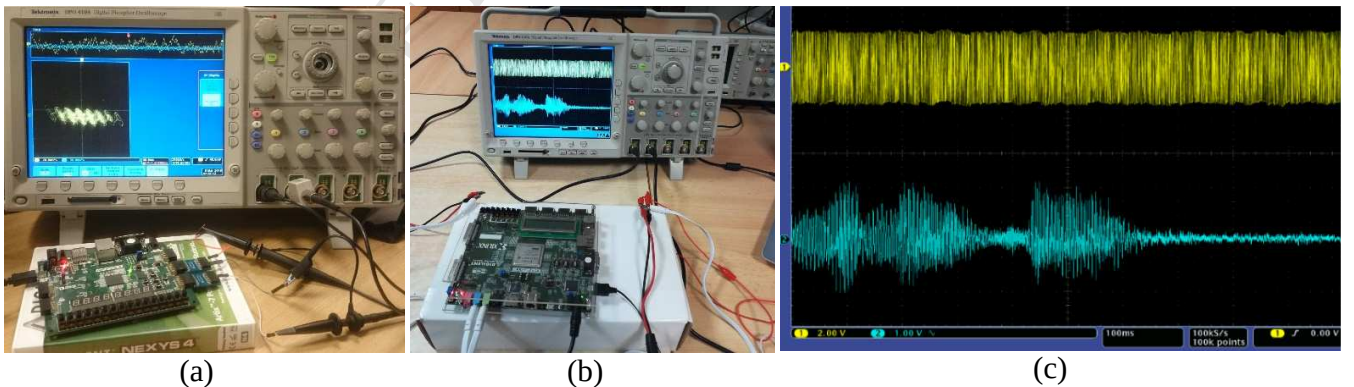


Fig. 12. a) oscilloscope setup of fractional chaotic systems, (b) oscilloscope setup of speech encryption and c) Speech encryption on FPGA, original and encrypted signals.

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